

TRUSTCHAIN
THIRD OPEN CALL FOR PROPOSALS

DESTINO

**Decentralized Ecosystems for Software Trustworthy
IntegratiOn**

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1. PROJECT SUMMARY

The Decentralized Ecosystems for Software Trustworthy INtegratiOn (DESTINO) project aims at defining, designing and implementing a robust, decentralized ecosystem for provable, evidence-based quality assurance of software, based on reputation and past behavior of developers. Modern software systems are rarely developed starting from scratch, on the contrary they are usually a patchwork of pre-existing libraries and modules glued together to achieve the desired goal. This approach, although capable of drastically reducing development time and up-front costs, exposes the products to quality problems that cannot be controlled easily by the integrator; in fact, the quality of the complete artifact is lead by the quality of its worst component and this may jeopardize the success of the project on many different aspects, ranging from cybersecurity, to user experience.

The DESTINO project aims at creating a public, immutable history of the endeavors of developers to generate an evidence-based reputation and allow integrators to have a better understanding of the skills involved in implementing the different building blocks and make better informed decisions.

2. APPLICANT BACKGROUND

The consortium includes two partners with complementary skills able to address all the challenges of the proposed project:

1. **Centre for Applied Software Engineering (CASE):** it is a SME focusing on providing services and tools related to software quality and helping companies in evaluating produced or acquired systems.
2. **University of Padua (UNIPD):** Dating back to 1222, the University of Padua (UNIPD) is one of the oldest and most prestigious seats of learning in Europe. The Department of Information Engineering (DEI), established in 1987, has about 110 tenured research staff, and about 200 between PhD students and Post-docs.

The team leaders of the involved organizations are the following:

- **Alberto Sillitti (CASE):** he received a Ph.D. degree in electrical and computer engineering from the University of Genoa, Italy, in 2005. He has been involved in several EU funded projects related to software quality, open source software, services architectures, and agile methods in which he applies noninvasive measurement approaches. Currently, he is co-founder, CEO, and chief scientist at the Centre for Applied Software Engineering (Italy). Previously, he was Full Professor with the Faculty of Computer Science and Engineering at the Innopolis University (Russian Federation) (2016-2022), where he led the Cyber-Physical Systems Lab and the Institute of Information Systems. He also served as Associate Dean for consulting activities. He authored more than 200 papers published in international conferences and journals. He has served as a member for the program committee of several international conferences, as Program Chair for OSS in 2007 and 2019, XP in 2010 and 2011, SEDA in 2012, 2013, and 2014, and as General Chair for SEDA in 2018 and OSS in 2020. In the context of the project, he will coordinate the overall project.
- **Mauro Migliardi (UNIPD):** Mauro Migliardi got a Laurea in Electronic Engineering and a PhD in Computer Engineering from the University of Genoa. He was a Research Associate and Assistant Professor at the University of Genoa and Research Associate at Emory University. Currently he is Associate Professor at the University of Padua. He is a member of the Scientific Committee of the Center for Computing Platforms Engineering,

Commentato [1]: (Maximum 300 words)

Commentato [1R2]: Ci siamo

Commentato [2]: (Maximum 1 page)

2. Team/consortium profile (If applicable, in case a team of natural persons/ consortium of legal entities apply)

-Describe the team, its main expertise and business area.

-List the members of the participating organisations that will directly work on the project (name, job title, main expertise & role in the project) and provide evidence of the main publications, projects, product/service portfolio, patents and relevant contributions of the different natural persons/organisations part of the team/consortium in line with your proposal.

-Describe the team/consortium partners' synergies and their relevance for the proposed project and TrustChain 3rd Call.

he has won the 2013 Canada-Italy Innovation Reward and he leads the joint CIPI-Gruppo Sigla s.r.l. Lab for CyberSecurity. He participated to several national and international research projects, recently the EU funded TRACE4EU Project, Topic IP DIGITAL-2022-DEPLOY-02-EBSI-SERVICES, Digital Europe Programme, where he is PI for the Università di Padova Unit, and the European NGI Open Call funded Project "Truth Seekers Chain (TSC)" (<https://www.trublo.eu/truth-seekers-chain/>) where he was PI/Scientific Coordinator. He has also chaired or co-chaired several international workshops and conferences and he is part of editorial boards. His main research interest is distributed systems engineering in general but recently he focused on SSI and DLT. He has tutored more than 100 among Bachelor, Master and PhD students at the Universities of Genoa, Padua and Emory, and he has authored or co-authored more than 150 scientific papers published in peer reviewed conferences, books and journals.

Overall, the skills of the team map as follows with the tasks of the project:

TABLE 1: Summary of tasks

Task	CASE	UNIPD
T1		X
T2	X	X
T3	X	X
T4	X	X
T5	X	
T6	X	

This is a short list of papers published by the team related to the topics of the project:

- Baldo T., Migliardi M., "SwarmAd: A decentralised content management system", IET Cyber-Physical Systems: Theory and Applications, 2023
- Guerar M., Migliardi M., TruthSeekers Chain: "Leveraging Invisible CAPTCHA, SSI and Blockchain to Combat Disinformation on Social Media", Lecture Notes in Computer Science, 2022, 13380 LNCS, pp. 419–431
- Guerar, M., Merlo, A., Migliardi, M., Palmieri, F., Verderame, L., "A Fraud-Resilient Blockchain-Based Solution for Invoice Financing", IEEE Transactions on Engineering Management, 2020, 67(4), pp. 1086–1098, 9020009
- Kapllani G., Khomyakov I., Mirgalimova R., Sillitti A., "An Empirical Analysis of the Maintainability Evolution of Open Source Systems", 16th International Conference on Open Source Systems (OSS 2020), Innopolis, Tatarstan, Russian Federation, 12 - 14 May 2020.
- Khomyakov I., Sillitti A., "A Novel Approach for Collecting and Sharing Software Metrics Data", 34th ACM Symposium on Applied Computing (SAC 2019), Limassol, Cyprus, 8 - 12 April 2019.

- Remencius T., Sillitti A., Succi G., "Assessment of software developed by a third-party: A case study and comparison", Information Sciences, Elsevier, Vol. 328, pp. 237 - 249, January 2016.

3. DETAILED PROPOSAL DESCRIPTION

3.1 CONCEPT AND OBJECTIVES

In the context of the DESTINO project, the goal is not to create a tokenomic based on the exchange of data, but to create a robust ecosystem for the building and management of provable, evidence-based reputation for software developers. Modern software development is based on the reuse and combination of modules previously developed. As such, the whole quality of the final product is heavily dependent on the lowest quality component, not just in terms of the quality of the actual code but also in terms of the capability of the original developers to maintain it. While the first aspect could be, in the case of Open Source software, measured on the actual component to be integrated, the second aspect can only be predicted. We claim that, for this reason, the availability of a robust, evidence-based reputation system is of paramount importance for the integration of software modules into quality, trustworthy systems.

The availability and immutability of the elements that constitute the reputation of a software developer (being it a single person, a group of developers or a company) will be guaranteed by the registration of such data onto a public, permissioned, blockchain based ledger.

The evidence of software quality and maintenance quality will be constituted by two different types of data: evidence of software quality for past developed projects and evidence of continuous and timed maintenance of developed software systems.

In the case of software quality, we plan to:

- 1) define (or select) one or more software quality metric
- 2) implement a tool capable of automatically measuring the metric on a source code
- 3) port the tool into the blockchain translating it into a smart contract to achieve transparency and trustworthiness of the procedure of measuring software quality and store that information for future reference.

In the case of continuous and timed maintenance of software systems, we plan to develop a smart contract that users and developers may leverage to report discovered bugs and vulnerabilities and to report the availability of solutions for those bugs and vulnerabilities.

Of course, there cannot be truth without accountability, hence all the information registered into the ledger will be signed using an eIDAS 2.0 compliant Qualified Electronic Signatures or, in the case of companies, seals.

Searching the ledger for all the information that constitute the reputation of a developer might be a cumbersome task, hence we also plan to enhance the usability of the reputation system by providing a search service based on a noSQL database constructed by collecting notifications of all the writes performed by our smart-contracts. This service would allow searching for a specific developer, but also to look for the past performance in developing a specific type of application or system.

3.2 PROPOSAL SOLUTION

Our solution aims at achieving the following goals

TABLE 2: Goals of the project

Commentato [3]: (Maximum 1 page)

Commentato [4]: -Describe the specific objectives of your proposal and explain the overall concept underpinning your proposed solution considering the TrustChain overall goals and specific OC3 objective on Economics and Democracy.

-It should be clear:

What are the needs and what are the relevant market trends?

What TrustChain OC3 challenges are you addressing and what is the methodology you are adopting?

What existing solutions (including your own) from industry and from the scientific literature partly address the challenges?

What is the human centric approach you are going to follow and how this is relevant for the work proposed

What new value proposition are you offering, to which target group?

What would be the benefits for TrustChain Large Scale Pilot.

The objective of Open Call #3 is to define and build mechanisms for smarter data exchange and data trading as well as innovative win-win federated business models' open data.

challenges:

- The current form of data-sharing practises does not fairly reward the data owners/content producers. Data platform owners make decisions around the terms and conditions of data sharing.

... [1]

Commentato [5]: • Enabling a sustainable circular economy that involves sharing, leasing, reusing existing materials and products demand data provenance on the transformation of properties of materials and on processes applied on them. Means to achieve such an economy requires innovation in terms of material traceability and digital passports.

- Marketplaces for AI/ML models trading/exchange

... [3]

Commentato [6]: • Accurate data discoverability on such marketplaces is a challenge. Precise matchmaking between sellers and buyers on a marketplace can significantly improve its performance.

- Data trading and exchange raise significant concerns about data privacy and security. There is a risk that sensitive or personal data could be mishandled, leading to privacy breaches, identity theft, or other malicious

... [2]

Commentato [7]: Mi chiedo, ma perche' fidarti della reputazione dello sviluppatore se hai un tool per effettuare una misura della qualita' del codice che ottieni? Allora ho pensato di aggiungere l'aspetto della capacita' di manutenzione, che non ha una misura diretta, ma vede nella reputazione dello sviluppatore un buon predittore.

Commentato [7R2]: Io la vedo anche come risparmio di risorse, è vero che posso analizzare il codice ma perchè perdere tempo a farlo se qualcuno lo certifica?

Commentato [8]: max 2 pages

ID	Title	Description
1	Increased trust	Increase level of trust in developers reputations as it's evidence-based and recorded on a blockchain.
2	Augment software reusability	Augment software modules description with visual cues related to the source as well as information related to the measured quality, the maintenance history, and all the links to directly access the details and evidence.
3	Software Projects as source for contents	Record information from multiple software projects in the blockchain.
4	Reputation Profile for developers	Build a single credible and permanent reputation profile for each developer.
5	Rewarding	Verifiable, evidence-based reputation building is a practical, economically exploitable reward for developers.
6	Open ecosystem implementation	Create an open platform for integrating new software quality metrics and third parties validation mechanisms.
7	Ease search and allow narrowing search scope	Create a sibling, noSQL based, system that will automatically collect all the evidence written in the ledger to perform searches based on reputation levels, developers PID and narrow the scope to specific types of system to be taken into account.

To the best of our knowledge, at present, there is no mechanism to evaluate the reputation of a developer beyond sifting through his/her internet presence and social media exposure. Our project will provide a one stop solution to look for collected, immutable, and easily provable facts about past experiences of a developer and, through the search engine augmentation provided by the sibling noSQL based system, will also allow creating views of developers experiences in specific fields.

In our solution we envision these major components:

- 1) the ledger;
- 2) the storage;
- 3) the search system.

The ledger will need to be publicly accessible for reading but permissioned to use smart contracts. The best match will be found in T1 and T2 also taking into account the evolution of European Initiatives such as EBSI and Alastria.

We do not plan to develop any form of storage. The storage will be indicated by the users, however, the type of storage will have an impact on the reputation building as adopting more robust solutions for storage will guarantee a better durability.

The search system will leverage an existing noSQL solution to guarantee high search throughput. We do not plan to develop a new noSQL system, the best match for our project will be identified in T1 and T2.

A developer willing to augment his reputation can add data in two different ways: adding a project or adding information about a maintenance activity of a previously added project.

In the first case, the developer will go through a procedure composed of these steps. First, he will collect his project in a single archive; second, he will add a Qualified Electronic Signature (eIDAS 2.0 compliant, QES henceforth) to the archive; third, he will store the archive into some publicly accessible storage system; fourth, he will prepare and sign (with the same QES as above) a

submission form that includes the hash of the archive and the link to retrieve it; fifth, he will contact a smart contract that will take as inputs the form and the archive. The smart contract will first, check the signature in the form; second, retrieve the archive using the link in the form; third, check that *i)* it has the same signature that is in the form and *ii)* it has the same hash that is in the form; fourth, extract the software from the project and measure the quality metrics of the project; fifth, write in the ledger the form, the measured metrics, and tags for the type of software system; sixth, signal the sibling system that there is a new data-point. Among the metrics written on the ledger, there will be also a reference to the persistence level of the storage used to make the project publicly available, in fact, in order to maximize the democratization of the procedure, it will be possible to submit projects that are simply available on the developer website; however, robustness of type of storage (from storage provided by services such as git-hub, to decentralized solutions such as IPFS) will enhance the score as a more robust storage will guarantee an additional level of transparency as a project won't simply "disappear" to prevent rechecking it or adding new metrics to its original evaluation.

In the second case, we envision the situation in which some user fills a "request for software maintenance" form, puts a Qualified Timestamp (eIDAS 2.0 compliant, QT henceforth) on it and signs it with a QES; then he sends it to the developer. At this point the developer will provide the maintenance patch and request a compliance declaration from the user. The compliance declaration will include a QT and the QES of the user and the hash of the patch. Now the developer may start another transaction to augment his reputation performing the following steps. First, he will collect the request for maintenance, the compliant declaration and the maintenance patch in a single archive; second, he will add his QES to the archive; third, he will store the archive into some publicly accessible storage system; fourth, he will prepare and sign (with the same QES as above) a submission form that includes the hash of the archive and the link to retrieve it; fifth, he will contact a smart contract that will take as inputs the form and the archive. The smart contract will first, check the signature in the form; second, retrieve the archive using the link in the form; third, check that *i)* it has the same signature that is in the form and *ii)* it has the same hash that is in the form; fourth, extract the request for maintenance, the declaration of compliance and the patch, check that all the signatures are correct and that the declaration of compliance refers to the provided patch; fifth, write in the ledger the form and the timeliness normalized over the complexity of the patch; sixth, signal the sibling system that there is a new data-point.

A user may search our sibling system either referencing a specific developer, or referencing a specific type of software, or specifying both. Such a search will provide accordingly information about the reputation of a single developer, the reputation of all developers that provided data about projects of the given type, or the reputation of that specific developer with reference only to that specific type of software projects.

3.3 EXPECTED IMPACT

Our project will contribute the objective and the challenges of the call addressing the following aspects:

"The objective of Open Call #3 is to define and build mechanisms for smarter data exchange and data trading as well as innovative win-win federated business models' open data."

Our project will develop a novel approach for developers reputation collection and management enabling transparency and democratizing the quality evaluation of software. In such a context, anybody can verify quality claims and track each component of a software system. The main idea

Commentato [9]: C'e' un modo di valutare una "complessita'" della patch? Magari una patch complessa richiede piu' tempo e quindi una risposta in un mese per una cosa complicata vale piu' di una risposta in un giorno per una cazzatella?

Commentato [9R2]: si potrebbero valutare diverse cose:
1) numero delle linee di codice aggiunte/modificate/rimosse
2) la modifica dei flussi decisionali
3) l'inserimento di nuove strutture (classi, metodi, ecc.) o la modifica di strutture esistenti
...

Commentato [10]: max 2 pages

is to democratize what some large companies do with their “clearing houses” that continuously inspect software components that are white listed for internal usage. This continuous quality assessment is very costly and requires specialized skills that only a few companies can afford.

“The current form of data-sharing practises does not fairly reward the data owners/content producers. Data platform owners make decisions around the terms and conditions of data sharing.”

Our platform allows data owners to take control of their data but at the same time users of such data are guaranteed that they can always use the data that they were granted access to. The reputation mechanism introduced will allow even small software producers to expose their commitment and effort in maintaining their products. In this way, it will be easier for integrators to select components based on objective criteria and not selecting only the ones made by well-known and established big players.

“Accurate data discoverability on such marketplaces is a challenge. Precise matchmaking between sellers and buyers on a marketplace can significantly improve its performance.”

Our platform also includes a synched mirror of the data stored in the blockchain to enable search inside the data keeping the possibility to verify them on the original blockchain.

Trust is a critical aspect of software since it is present in every aspect of our life. Moreover, since systems include an enormous amount of different components from different sources, it is more and more difficult to really understand and trust them. Our approach provides a way to trace the reputation of the developers of each component and have a better understanding of the resulting product.

The impact on the software industry is clear: it will be easy to understand if a component can be used as part of a larger system and which is its impact on the overall system. Moreover, end-users can verify the reputation of the developers of each component enabling a clear picture of the overall quality of a software system. In this way, there will be an increased level of transparency pushing towards a global improvement of software quality. Since software is now part of all industries, it is clear the potential impact that the project might have.

The business of CASE is focused on software quality. In particular, it provides consultancy services and tools to analyze the quality of software from the points of view of the development processes and the code produced. It provides these services to both software producers (mainly to help them in identifying problems and improve the overall quality) and to software consumers (mainly to assess the software they are acquiring to guarantee the maintainability in the long term). For this reason, the project is perfectly aligned with the business focus. In particular, CASE will benefit from this project in a number of different ways:

1. improved visibility in the software quality community
2. the possibility to develop new value-added services and tools based on the information available in the blockchain and focused on specific application domains to help integrators to select the best fitting components for their needs.

As a high education organization, the business of UNIPD is mainly related to education, research, and technology transfer. The project will affect all these aspects. About research, the activity performed in this project can be classified as applied research that will be disseminated through scientific papers. About education, the knowledge developed in the project will be disseminated to MSc and PhD students through seminars and involvement into the project. About technology

transfer, the project aims at developing a sustainable business based on solid scientific outcomes that can be managed by the partner company.

The results of this project will be disseminated and communicated using different means, including:

1. **Web site:** it will be developed during the project to collect and make available the early results and implementations.
2. **Community building:** we will focus on the development of a community of users that will benefit from the usage of the platform. We will focus on both producers of components and integrators.
3. **Papers:** we will produce a number of scientific papers dealing with the developed system.

3.4 BUSINESS MODEL AND SUSTAINABILITY

The project aims at developing a new ecosystem where integrators can base their decision of using a specific component assessing the reputation of a developer without an extensive amount of specialized investigation.

Currently, some large corporations have developed internal “clearing houses” that perform an evaluation of software components that can be used by the company. Performing such analysis and keeping the catalog up-to-date can be very expensive and only very large companies can afford, hence, most SMEs cannot afford such infrastructure due to lack of specific skills needed and/or their cost. Therefore, they do not perform such a deep check for every component they use, increasing the related risks. Moreover, often they select by default components from large and well-known organizations avoiding smaller and less-known producers because of their lack of deep knowledge about the specific organization. In such cases, the selection is performed based on the a-priori perception without any objective information. This approach is focused on minimizing the perceived risk of using a component but it is almost ignoring the features offered by the component. This approach results in selecting sub-optimal components just because of the lack of information. Our system will support integrators in an objective evaluation of the best components fitting their project enabling an optimal and non-biased selection.

The business model is based on the democratization of such “clearing house” enabling developers to expose their reputation into a public ledger where everybody can verify their reputation at any given time. Since the blockchain is designed to store transaction information reliably but it provides only basic support in the retrieval of the information, an advanced search system is required. Moreover, a number of value-added services and tools can be developed on the basic infrastructure to cope with the needs of companies of different sizes and different application domains.

Possible added-value services include the following:

- A reliable (decentralized) storage of the evaluated components to improve the level of trust and the replicability of the analysis.
- Enterprise software catalog management to keep track of the components that can be used in a single enterprise, shared with all the employees, and updated whenever new versions are available and/or the reputation of the developers change.
- Certification of the reputation of closed-source components. In this case, the code is not public and there is the need of a trusted third-party (certifier) that is entitled to look at the code and perform all the operations to calculate and update the reputation.

3.5 IMPLEMENTATION

The project is organized in 5 technical tasks and 1 administrative:

T1: State of the art

Having a full awareness of the state of the art is a fundamental preliminary activity for any project. In order to complete this activity, we will analyze the available literature leveraging databases such as IEEEExplore, Scopus, ScienceDirect and the ACM library; besides, we will also monitor the evolution of related European initiatives (such as EBSI) and regulation (eIDAS2.0). The synthesis of such analysis will constitute deliverable D1.

The task will be led by UNIPD.

T2: Requirements and specifications

Based on the deep analysis of the state of the art and extending the idea presented in this proposal, a set of detailed requirements and specifications will be generated. The development will be based on agile methods (SCRUM in particular), therefore at the beginning only a high level description will be available (deliverable D2) but at every iteration the details will be investigated in depth and the plan will be adapted accordingly.

The task will be led by UNIPD.

T3: Implementation

Due to the limited timeframe available, the implementation of the proposed system will include as much as possible already existing open source components. For this reason, we plan to adopt a micro-service architecture to be able to integrate subsystems using different technology stacks. Moreover, the development approach will be based on Agile (SCRUM in particular) to provide continuously valuable results and reduce the risks related to a single large release at the end of the project. The definition of the initial implementation roadmap (D3.1) will begin as soon as a first set of requirements will be available, while the documentation (D3.3) will be created along the development of the system (D3.2) and it will be available at the end of the project.

The task will be led by CASE.

T4: Validation

The validation of the system will be very limited due to the time constraints of the project. For this reason we plan to validate the different components when they will be available. In the validation roadmap (D4.1) we will identify when the different subsystems can be tested (also based on the development roadmap defined in D3.1). Therefore, the actual validation will happen during almost the entire project, while the overall results will be available at the end in the D4.2.

The task will be led by UNIPD.

T5: Business model

A very high level draft of the business model has been defined in section 3.4, however it is just a starting point. A more detailed model and a related plan will be defined during the development of the project (D5.1 and D5.2). In particular, a deep analysis of the different possibilities and the related sustainability will be performed at the beginning of the project. Since the product-market fit is extremely difficult to achieve at the first attempt and in such a short time, we plan to identify different possibilities and approaches that can be implemented into the system, limiting as much as possible the impact on the implementation. In this way, we will be able to explore different possibilities and adopt the ones that produce the best results.

Commentato [11]: (Maximum 2 pages)

- Provide an overview of your overall work plan considering the 9 months' timeframe of TrustChain Open Call #3.
- Analyse the activities you plan to execute in each task and the functionality/services you are going to deliver: objective, duration, implementation steps, resources available. Illustrate the timing of your activities using a Gantt diagram or similar. The co-creation approach should be made evident. Please note that if relevant seeking and obtaining the mandatory approvals in a timely manner from competent ethics and/or data protection bodies are the applicant's sole and exclusive responsibility and that the absence of such approvals, when and where legally required, may void the eligibility of the applicant's proposal.
- Outline a relevant risk management plan.
- Outline the management procedures and overall resources you are aiming to employ.
- Use the table hereafter in order to help you present the requested information.

The task will be led by CASE.

T6: Project management

The task will focus on the coordination of the different activities performed in the project and the tracking of the progress according to the proposed plan. Moreover, proper actions will be taken if deviations will appear or risks will materialize. Finally, the activity will include the collection of all costs statements and the creation of expected reports.

The task will be led by CASE.

Figure 1 summarizes the tasks organization and their relationships, while Figure 2 shows the Gantt diagram of the tasks.

FIGURE 1: Tasks organization

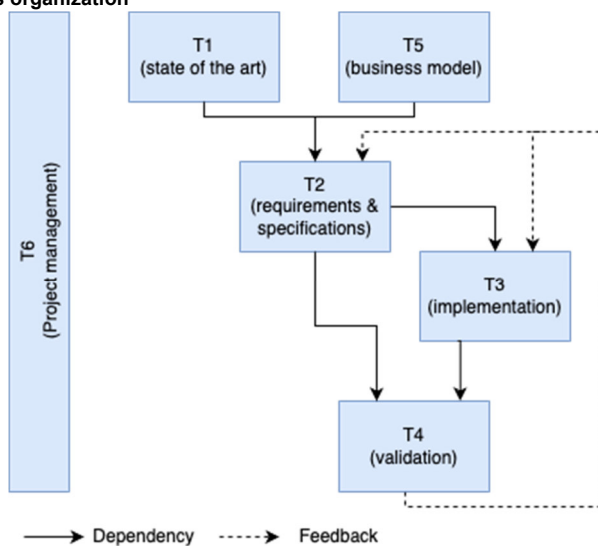


FIGURE 2: Summary of tasks

Task	M1	M2	M3	M4	M5	M6	M7	M8	M9
T1									
T2									
T3									
T4									
T5									

T6									
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3.5.1 Deliverables and milestones

TABLE 3: List of deliverables

No.	Deliverable or milestone name	Description	Type	Delivery Month	TRL level delivered
D1	Deliverable	State of the art	Report	1	
D2	Deliverable	Technical specification	Report	2	
D3.1	Deliverable	Implementation roadmap	Report	2	
D3.2	Deliverable	Implementation	Software	8	7
D3.3	Deliverable	Documentation	Report	9	
D4.1	Deliverable	Validation roadmap	Report	4	
D4.2	Deliverable	Validation results	Report	9	
D5.1	Deliverable	Preliminary business plan	Report	3	
D5.2	Deliverable	Final business plan	Report	9	

The T6 will be in charge of the project management activities, progress monitoring, and risks management during the lifetime of the project. In particular, we were able to identify the following risks:

TABLE 4: Risks

No.	Risk	Likelihood	Mitigation plan
R1	Failure to deliver the final system	Low	The implementation plan will be designed incrementally to avoid a single delivery at the end of the project. Even if some iteration may fail in achieving the expected results, we expect to have a working system with some reduced functionalities.
R2	Failure to define the proper technologies to use	Medium	Many different approaches can be selected from the technical point of view and the selection of the wrong technology may affect the entire project. However, since we do not expect to develop new technologies in the limited timeframe of the project, we expect to be able to change components if we realize that the selected ones are not fitting properly.
R3	Failure to be financially sustainable	High	Since achieving product-market fit is difficult in a short time, the business model and the related plan take into consideration different options that will be explored during the project and afterwards. This approach will increase the possibility of finding the proper approach.
R4	Failure to involve developers and/or integrators	High	We are aware that promoting the project in the communities is important for the long term success. For this reason we plan to disseminate the results as soon as possible and to build and keep up-to-date a website promoting the project. However, most of the marketing of the system will be performed after the end of the project.

Commentato [12]: Please add a list of deliverables and milestones using the table provided. 4 deliverables are mandatory for TrustChain, please consider them in the list of deliverables (e.g., documents, reports, user manual, a tool ...) you intend to submit. Justify each of them with a small description and state the relevant TRL level for each deliverable.

-Describe the specific objectives of your proposal and explain the overall concept underpinning your proposed solution considering the TrustChain overall goals and specific OC3 objective on Economics and Democracy.

-It should be clear:

What are the needs and what are the relevant market trends?

What TrustChain OC3 challenges are you addressing and what is the methodology you are adopting?

What existing solutions (including your own) from industry and from the scientific literature partly address the challenges?

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What new value proposition are you offering, to which target group?

What would be the benefits for TrustChain Large Scale Pilot.

The objective of Open Call #3 is to define and build mechanisms for smarter data exchange and data trading as well as innovative win-win federated business models' open data.

challenges:

- The current form of data-sharing practises does not fairly reward the data owners/content producers. Data platform owners make decisions around the terms and conditions of data sharing.
- Accurate data discoverability on such marketplaces is a challenge. Precise matchmaking between sellers and buyers on a marketplace can significantly improve its performance.
- Data trading and exchange raise significant concerns about data privacy and security. There is a risk that sensitive or personal data could be mishandled, leading to privacy breaches, identity theft, or other malicious activities.< no personal data>
- Establishing enforceable data marketplace contracts for data exchange is missing; Clear and fair service level agreements for both sellers and buyers needs to be in place for a trustworthy marketplace.
- There is no standard pricing model for data, making it challenging to determine the fair market value of different data types. Mechanisms to ascertain the data quality and hence its price fairly is a challenging issue as the value of data can be subjective and context-dep

- Accurate data discoverability on such marketplaces is a challenge. Precise matchmaking between sellers and buyers on a marketplace can significantly improve its performance.
- Data trading and exchange raise significant concerns about data privacy and security. There is a risk that sensitive or personal data could be mishandled, leading to privacy breaches, identity theft, or other malicious activities.< no personal data>
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- Issue around data provenance exist as it can be difficult to verify the data source. Techniques to inspect the provenance of a specific product, service or data is missing. Open reputation management can be one of the solutions, but it requires careful design considerations.<digital signature of developer and metrics extraction>
- Establishing clear data governance practices, including data access controls and usage

policies, is essential for responsible data exchange. Data governance frameworks can be complex to implement and enforce.

- Federated marketplaces among self-interested parties emerge in various contexts, such as cloud services, IoT data exchange and more, for increasing service coverage, availability, efficiency, etc. Providing support for trustworthy service provision logging, transparent billing, fair value sharing and coordination in resource allocation in these contexts is a challenge.
- Innovative incentive mechanisms for decentralized, coordinated outcomes, potentially involving tokenomics, in various application contexts should be provided.

- Enabling a sustainable circular economy that involves sharing, leasing, reusing existing materials and products demand data provenance on the transformation of properties of materials and on processes applied on them. Means to achieve such an economy requires innovation in terms of material traceability and digital passports.
- Marketplaces for AI/ML models trading/exchange has witnessed a rise with advancement in AI training models and fear of data exchange due to privacy violations. A comprehensive comparison of such models on marketplace remains a challenge.
- Digital solutions to enhance civic mobilisation and engagement. Such solutions can utilize real-time information, platforms for crowdsourcing ideas, and platforms for facilitating access to participatory channels, in order to transform and enhance democratic decision-making.